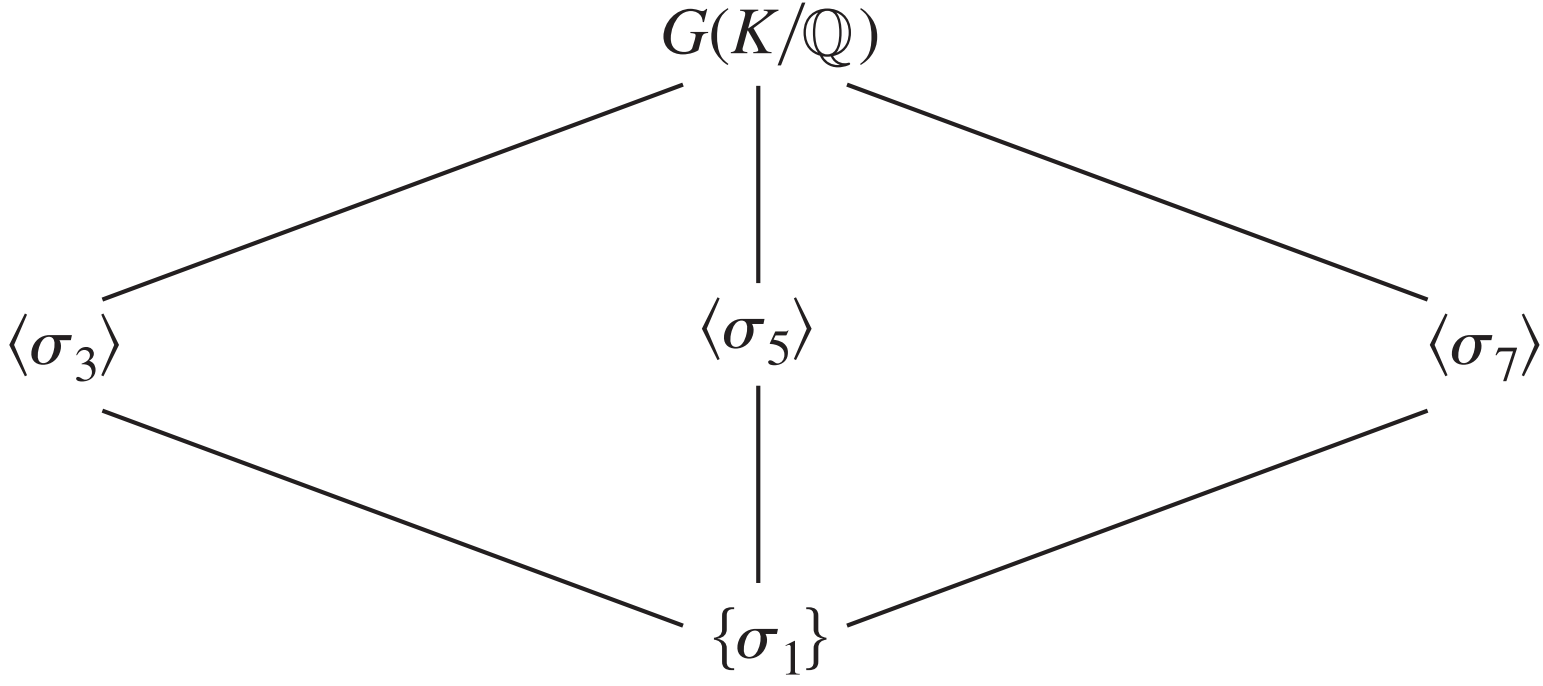

$G(K/\mathbb{Q})$
$\langle\sigma_3\rangle$
$\langle\sigma_5\rangle$
$\langle\sigma_7\rangle$
$\{\sigma_1\}$
(a)